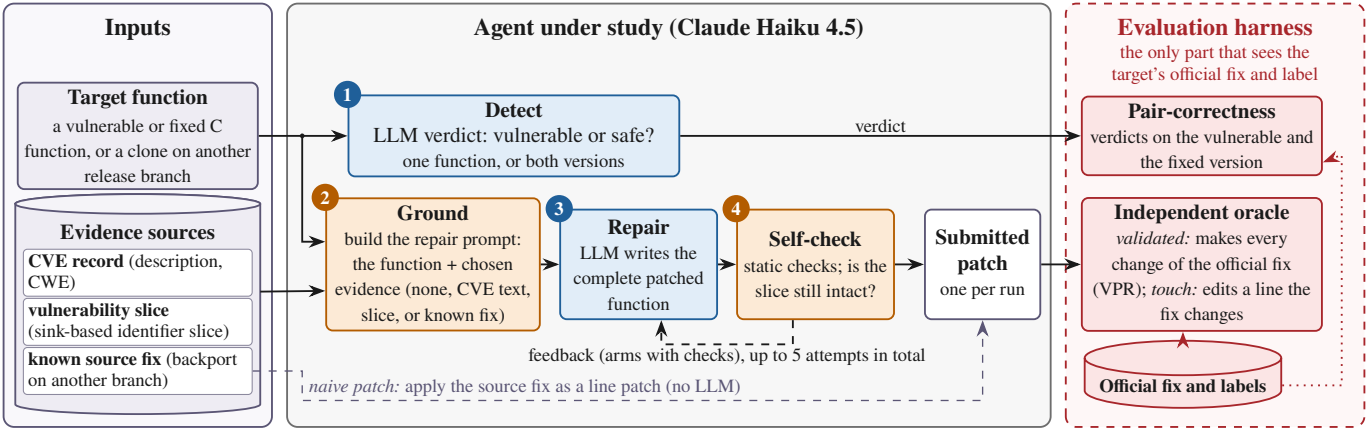



**RQ1 Grounding**

**Varies:** evidence in 2 and checks in 4 (ladder A0–A4)

**Measures:** VPR, touch

**Data:** 150 Big-Vul CVEs

**RQ2 Detection**

**Varies:** prompts, reasoning, models in 1 (Haiku 4.5; Qwen2.5-Coder 7B/14B)

**Measures:** pair-correct

**Data:** 150 CVE pairs (94 pre-, 56 post-cutoff)

**RQ3 Contamination**

**Varies:** the inputs: pre/post-cutoff CVEs, certified slice-preserving variants, likelihood probe

**Measures:** accuracy change vs. lookup controls

**Data:** 217 matched functions; repair: 150 + 56 CVEs

**RQ4 Porting**

**Varies:** whether 2 gets the known source fix (P0–P4)

**Measures:** VPR

**Data:** 88 hard clones, 23 CVEs